

DFIR Pre/Post-Breach Package Comparison

Host: datafiles-corp-s01 — Cyber Range Capstone honeypot lab

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1. Baseline Determination

Baseline package	PreBreach_MDE_Investigation_Packg.zip ("PRE")
Baseline collected	2026-09-01T16:01:06Z – 16:01:27Z (per Forensics Collection Summary.csv)
Comparison package	AfterBreach_MDE_Investigation_Packg.zip ("POST")
Comparison collected	2026-09-03T01:49:42Z – 01:50:05Z (per Forensics Collection Summary.csv)
Elapsed time	~34 hours between collections

PRE is treated as the baseline. Its collection timestamp sits ~28 minutes after the honeypot's recorded exposure time (2026-09-01T15:32:42Z, per the build checklist) and ~4h26m before the first confirmed attacker activity against this host (the MySQL ransom event documented in the incident report, ~20:27 UTC the same day) — it captures the system in its as-hardened, freshly-exposed state. POST was collected within ~30 minutes of the recorded isolation timestamp (2026-09-03T02:16:28Z), representing the system's state at containment. Both packages share identical MDE live-response folder structure, confirming they are directly comparable collections from the same host.

2. Summary Verdict

Compromise indicators found	YES — attempted/in-progress credential access; NOT a confirmed successful intrusion
Confidence	High that brute-force attack activity occurred against this host; Medium-Low that it materially compromised it

The Security event log shows 22,906 failed logon attempts (Event ID 4625) and 110 account lockouts (Event ID 4740, targeting the built-in Guest account) from at least 11 external IP addresses absent from the PRE baseline, including one IP (195.201.175.13, responsible for 66% of the failures) that still held two live TCP connections to port 3389 at the moment of the isolation snapshot. Against that, every other artifact category compared below — persistence/autoruns, services, scheduled tasks, local accounts and group membership, running processes, installed programs, DNS cache, ARP, and firewall rules — is unchanged or explainable by the simple absence of a logged-on interactive user at POST collection time. No successful authenticated logon (interactive or RDP) was found from any external IP in either package.

3. Notable Deltas

Category	Item	Change	Assessment	Evidence
Security Event Log	22,906x Event ID 4625 (failed logon, LogonType 3/NTLM) from 11 external IPs; top source 195.201.175.13 = 15,204 (66%)	ADDED (0 in PRE)	Suspicious	post/Security Event Log/Security.evtx; 2026-09-02T14:13:30Z–2026-09-03T01:49:45Z
Security Event Log	110x Event ID 4740 (account logout), TargetUserName=Guest (SID ...-501)	ADDED (0 in PRE)	Suspicious (effect of #1)	post/Security Event Log/Security.evtx
Network Connections	2x ESTABLISHED TCP 10.2.0.23:3389 ↔ 195.201.175.13	ADDED	Suspicious	post/Network Connections/ActiveNetConnections.txt
Security Event Log	3x successful Event ID 4624 "ANONYMOUS LOGON" (Type 3/Network) from 34.156.47.23 (1) and 111.53.72.251 (2, also a brute-force IP)	ADDED	Suspicious	post/Security Event Log/Security.evtx
Security Event Log	0 successful LogonType 10 (RemoteInteractive/RDP) events from any external IP, in either package	UNCHANGED	Benign — no completed RDP compromise	Security.evtx, EventID 4624 LogonType field
Autoruns / Run keys	HKLM Run key byte-identical (SecurityHealth only); HKU\500 Run key absent in POST	UNCHANGED / explainable	Benign	Autoruns.txt; no loaded user hive at POST (QueryUser.txt: "No User exists for *")
Services (registry)	716 normalized service keys, 0 added/0 removed; 27 ImagePath diffs = string-format noise or Defender platform bump (4.18.26070.9→4.18.26080.3)	UNCHANGED	Benign	Autoruns.txt Services block, normalized diff
Services (CIM)	24 per-user session services (Bluetooth, Clipboard, PenService, PrintWorkflow, etc.) absent in POST	REMOVED / explainable	Benign	Services.csv; consistent with no logged-on user at POST
Scheduled Tasks	\Microsoft\Windows\Setup\SetupCleanupTask removed	REMOVED	Benign — built-in cleanup task	ScheduledTasks.csv diff; corroborated by single Event ID 4699 at 2026-09-02T15:12:49Z
Local Users & Groups	LocalGroups.txt byte-identical; no 4720/4732/4728/4722/4724/4738 events in POST	UNCHANGED	Benign — no new/privileged accounts	LocalGroups.txt; Security.evtx Event ID frequency table
Processes	26 process names present only in PRE (explorer.exe, msedge.exe, MySQLWorkbench.exe, rdpclip.exe, ...); 0 new process names in POST	REMOVED / explainable	Benign — reflects no interactive session, not process manipulation	Processes.csv diff
Prefetch	13 new .pf entries, all Defender/Windows-Update binaries (MsMpEng, NisSrv, MPCmdRun, SIHClient, WindowsPackageManagerServer, ...)	ADDED	Benign	Prefetch Files/Prefetch directory listing diff
Installed Programs	No differences	UNCHANGED	Benign	InstalledPrograms.csv + Uninstall registry exports
DNS Cache / ARP / SMB / IpConfig	1 stale Microsoft delivery-optimization DNS record swapped for 2 similar ones; DHCP lease renewed; ARP and SMB Session identical	ADDED/CHANGED	Benign — routine OS/network churn	DnsCache.txt, Arp.txt, SMB Summary.txt, IpConfig.txt

Category	Item	Change	Assessment	Evidence
Firewall/audit-policy events	Event IDs 4945/4946/4948/4957 (firewall rule) and 4717/4718/4719 (audit policy) present in PRE, entirely absent in POST	Not present in POST window	N/A — nothing to compare	Security.evtx Event ID frequency table
Rights assignment	136x Event ID 6426 removing SeDenyNetworkLogonRight/SeDenyRemoteInteractiveLogonRight for 2 SIDs whose domain identifier does not match this host's domain SID	ADDED	Unresolved — not classified	Security.evtx, EventID 6426, 2026-09-02T19:38:12Z
Defender ACL activity	12x Event ID 4907, MsSense.exe adjusting ACLs on its own investigation-collection temp folder	ADDED	Benign — self-generated by MDE collector	Security.evtx, EventID 4907, ~2026-09-02T15:45:16Z

4. Prioritized Indicators (MITRE ATT&CK)

#	Indicator	ATT&CK Technique	Confidence
1	Distributed/multi-source RDP+NTLM brute force: 22,906 failed logons, 11 external IPs, targeting common usernames (admin, administrator, remote, test, support, server, user, DataFiles)	T1110.001/.003 Brute Force (Password Guessing/Spraying); T1021.001 Remote Services: RDP	High
2	Top source 195.201.175.13 (66% of failures) held 2 live ESTABLISHED TCP sessions to port 3389 at isolation, with no matching successful 4624 LogonType 10 event	T1021.001 (attempted); inconclusive on intent/state of the session	Medium
3	Guest account locked out 110x as a direct consequence of #1	T1078.001 Valid Accounts: Default Accounts (targeted, not compromised)	High (as consequence, not independent activity)
4	3 successful ANONYMOUS LOGON (unauthenticated NTLM/SMB) sessions from 2 of the brute-force IPs	T1046 Network Service Discovery / T1135 Network Share Discovery	Low-Medium — no evidence of what was accessed

5. Probable Incident Type

External, opportunistic credential-access activity against the exposed RDP service (port 3389), with incidental anonymous SMB enumeration from two of the same source IPs. This mirrors the same internet-wide scanning pattern already documented against this host's exposed MySQL service (port 3306) in the primary incident report, arriving via a different vector roughly 18 hours later. No artifact in either package supports classifying this as a successful intrusion, ransomware, or data theft: no new accounts, no new persistence mechanism, no new services, no new processes, no successful authenticated interactive or RDP logon, and no installed-software changes were found. If a single label is required: attempted brute-force / credential access, uncompromised as of the POST collection.

6. Gaps

- Security.txt (the wevtutil text export) is 0 bytes in both packages — the evtx parse below could not be cross-validated against a human-readable export.
- Security.evtx is a fixed-size ring buffer (both files are exactly 21,041,152 bytes). The 22,906-event 4625 flood may have evicted earlier or lower-frequency event types before the true onset of the brute-force window — e.g., Event ID 4688 (process creation) appears 35x in PRE but 0x in POST; this cannot be distinguished from "no processes were created" using this data alone.

- Neither package includes System.evtx or Application.evtx — service crashes, application crashes, and other non-Security indicators could not be checked.
- No dedicated Windows Defender detection/threat log is present. MpCmdRunLog.txt shows only routine collector activity; it neither confirms nor rules out an AV detection during the window.
- Neither package includes a hosts file, browser artifact, or Credential Manager export — local credential harvesting cannot be ruled in or out from this data.
- Event ID 6426 (136 occurrences, foreign-domain SIDs) is not resolved to a specific cause; flagged in Section 3 but not conclusively classified as benign or suspicious.
- Neither package contains Event ID 5140/5145 (network share access) — what, if anything, the 3 ANONYMOUS LOGON sessions accessed could not be determined.

Prepared as a comparison appendix to the MS Defender/Sentinel incident report for datafiles-corp-s01. Findings reflect only what is present in the two supplied MDE live-response packages; items above are marked "Not determined from available logs" implicitly wherever the Gaps section applies.